

Problem set — Module 0 (Foundations)

Foundations only stick if you *use* them. These are short and mechanical on purpose — the goal is fluency, not cleverness. The two that matter most for what's coming: **P4** (telescoping) and **P8** (linearity of expectation). Solutions at the bottom.

Problems

P1 (sum basics). Evaluate, using closed forms (not by adding term-by-term): (a) $\sum_{k=1}^{100} k$ (b) $\sum_{k=1}^n (2k - 1)$

(c) $\sum_{k=0}^{10} 2^k$.

P2 (reindexing). Rewrite $\sum_{k=3}^n a_k$ as a sum whose index starts at 0. Then check your shift on $\sum_{k=3}^5 k$ by evaluating both forms.

P3 (splitting a range). Given $\sum_{k=1}^{2n} k = \frac{2n(2n+1)}{2}$, use range-splitting to find $\sum_{k=n+1}^{2n} k$ (the "second half"). Simplify.

P4 (telescoping — important). Evaluate $\sum_{k=1}^n \left[\frac{1}{k} - \frac{1}{k+1} \right]$ by telescoping. Then explain in one line how this is the same maneuver as "divide by $N(N+1)$ " in the Module 1 quicksort solve.

P5 (permutations vs combinations). (a) How many ways to seat 6 people in a row? (b) How many ways to choose a committee of 3 from those 6? (c) How many ways to choose 3 of the 6 and assign them president/VP/treasurer? State which rule (permutation or combination) each uses and why.

P6 (binomial coefficients). (a) Compute $\binom{6}{2}$ and $\binom{6}{4}$; confirm the symmetry identity. (b) Verify Pascal's rule for $\binom{5}{2}$: show it equals $\binom{4}{1} + \binom{4}{2}$. (c) Use the binomial theorem to expand $(x+1)^4$.

P7 (uniform probability). Roll two fair six-sided dice. (a) Size of the sample space? (b) $P(\text{sum} = 7)$? (c) $P(\text{sum} = 12)$? (d) $P(\text{the two dice show different values})$?

P8 (linearity of expectation — important). (a) In a uniformly random permutation of $1, \dots, n$, let X be the number of fixed points. Using indicators, show $E[X] = 1$. (Reproduce the notes' argument from scratch.) (b) Now the payoff shape: let Y be the number of **inversions** — pairs (i, j) with $i < j$ but the element at position i is larger than the one at position j . Using one indicator per pair and linearity, show $E[Y] = \frac{1}{2} \binom{n}{2} = \frac{n(n-1)}{4}$. (Hint: for a random pair, which order is it in, and with what probability?)

P9 (harmonic numbers). (a) Compute H_5 as an exact fraction. (b) Estimate $H_{1,000,000}$ using $H_n \approx \ln n + \gamma$. (c) True or false: H_n is bounded above as $n \rightarrow \infty$. Justify.

P10 (logs & Big-O). (a) Write $\lg n$ in terms of $\ln n$ (give the constant). (b) Explain in one sentence why " $f = O(n \log n)$ " doesn't tell you the base of the log, but " $f \sim 2n \ln n$ " does tell you everything. (c) Which is a stronger statement about f : $f \sim n^2$, or $f = O(n^2)$?

Solutions

P1. (a) $\sum k = \frac{100 \cdot 101}{2} = 5050$. (b) $\sum_{k=1}^n (2k-1) = 2 \sum k - \sum 1 = 2 \cdot \frac{n(n+1)}{2} - n = n(n+1) - n = n^2$.

(The sum of the first n odd numbers is n^2 — a classic.) (c) Geometric with $r = 2$:

$$\sum_{k=0}^{10} 2^k = \frac{2^{11} - 1}{2 - 1} = 2048 - 1 = 2047.$$

P2. Let $j = k - 3$ (so $k = j + 3$); as k runs $3, \dots, n$, j runs $0, \dots, n - 3$: $\sum_{k=3}^n a_k = \sum_{j=0}^{n-3} a_{j+3}$. Check on $a_k = k$, $n = 5$: original $3 + 4 + 5 = 12$; shifted $\sum_{j=0}^2 (j+3) = 3 + 4 + 5 = 12$. ✓

P3.

$$\sum_{k=n+1}^{2n} k = \sum_{k=1}^{2n} k - \sum_{k=1}^n k = \frac{2n(2n+1)}{2} - \frac{n(n+1)}{2} = \frac{(4n^2 + 2n) - (n^2 + n)}{2} = \frac{3n^2 + n}{2} = \frac{n(3n+1)}{2}.$$

P4. Telescoping: consecutive terms cancel, leaving first minus last:

$$\sum_{k=1}^n \left[\frac{1}{k} - \frac{1}{k+1} \right] = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

Same maneuver as Module 1: dividing the quicksort recurrence by $N(N+1)$ rewrote it as $T_N - T_{N-1} = \frac{2}{N+1}$, i.e. a difference of consecutive terms — so summing telescoped it down to a harmonic sum. Telescoping = "make it a difference, then collapse."

P5. (a) Order matters (a row) $\rightarrow$ permutation of all $6 = 6! = 720$. (b) Order doesn't matter (a plain committee) $\rightarrow$ combination $= \binom{6}{3} = 20$. (c) Choose 3 *and* order them into distinct roles $\rightarrow$ ordered selection $= 6 \cdot 5 \cdot 4 = 120$ ($= \binom{6}{3} \cdot 3! = 20 \cdot 6$). Equivalent view: pick the committee (20), then assign 3 roles ($3! = 6$).

P6. (a) $\binom{6}{2} = \frac{6!}{2!4!} = 15$; $\binom{6}{4} = \frac{6!}{4!2!} = 15$. Equal ✓ (symmetry: $k = 2$ vs $n - k = 4$). (b) $\binom{5}{2} = 10$; $\binom{4}{1} + \binom{4}{2} = 4 + 6 = 10$. ✓ (c) $(x+1)^4 = \sum_k \binom{4}{k} x^k = x^4 + 4x^3 + 6x^2 + 4x + 1$ (coefficients = Pascal row $n = 4$).

P7. (a) $|\Omega| = 6 \cdot 6 = 36$ equally likely ordered pairs. (b) Sum 7: $(1,6)(2,5)(3,4)(4,3)(5,2)(6,1) = 6$ outcomes $\rightarrow \frac{6}{36} = \frac{1}{6}$. (c) Sum 12: only $(6,6) = 1 \rightarrow \frac{1}{36}$. (d) Different values: complement of "same" (6 doubles), so $\frac{36-6}{36} = \frac{30}{36} = \frac{5}{6}$.

P8. (a) $X = \sum_{i=1}^n \mathbf{1}_i$, $\mathbf{1}_i = [\text{item } i \text{ is fixed}]$. Each item is equally likely to occupy any of the n positions, so $P(\mathbf{1}_i = 1) = \frac{1}{n}$. Linearity: $E[X] = \sum \frac{1}{n} = 1$ — independent of n . (b) $Y = \sum_i \{i$

P9. (a) $H_5 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{60+30+20+15+12}{60} = \frac{137}{60} \approx 2.2833$. (b) $H_{10^6} \approx \ln(10^6) + 0.5772 = 6 \ln 10 + 0.5772 \approx 13.8155 + 0.5772 = 14.39$. (c) **False** — H_n is unbounded; it grows like $\ln n \rightarrow \infty$ (just very slowly). The terms $\frac{1}{k}$ shrink to 0, but their partial sums still diverge.

P10. (a) $\lg n = \frac{\ln n}{\ln 2} = \frac{1}{\ln 2} \ln n \approx 1.4427 \ln n$. (b) Changing log base only multiplies by a constant, and $O(\cdot)$ already absorbs constant factors — so $O(n \log n)$ is *base-agnostic*. But " $\sim 2n \ln n$ " fixes the growth, the base (e), *and* the leading constant (2), which is the whole point of chasing $\sim$ rather than O . (c) $f \sim n^2$ **is stronger**. " $f \sim n^2$ " pins f to n^2 up to a vanishing relative error (it even fixes the constant to 1); " $f = O(n^2)$ " only caps f from above (f could be much smaller, even $\sim n$). Asymptotic equivalence $\Rightarrow$ big-O, not the reverse.

Next

That's the toolkit. Head to Module 1 — when the quicksort analysis pulls a $\frac{1}{N}$ probability, a telescoping sum, or an H_N out of the air, you'll now see exactly where each comes from.